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Famous Astronomers: Galileo Galilei

The First Telescope

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Introduction

This volume's subject is a person rather than an instrument, the series' one biography in the astronomical canon where the person and the instrument are inseparable: Galileo Galilei, the mathematician of Pisa and Padua and Florence, the man who did not invent the telescope but pointed it at the sky with a mathematician's discipline and told the world what he saw. The shelf's other volumes collect observatories and phenomena; this one collects a working life, sixty-eight years across the turn of the seventeenth century, in which the ancient sky of unchanging spheres was dismantled observation by observation: the Moon revealed as a rough earthlike world, the Milky Way resolved into uncountable stars, four small planets circling Jupiter, the phases of Venus, the spots on the Sun, all reported in a five-year flood of pamphlets and books that began in sixteen ten and broke the Aristotelian cosmos beyond repair. The chapters that follow treat the life as the series treats its subjects, as a complete story of an idea and its machinery: the man's formation, the instrument's arrival, the discoveries' sequence, the physics that underpinned them, the conflict they provoked, the trial that ended the public career, and the legacy the modern sky still carries.

The biography's shape is the series' familiar three acts. The first act is the obscure professor: the Pisan-born mathematician who failed his medical degree, taught geometry for bread, questioned Aristotle in a chair the university still held sacred, and found, in the proportional compass and the falling-body experiments, a physics that would not be published for decades. The second act is the discoverer: the Paduan professor who heard of a Dutch spyglass, worked out the optics himself, built better instruments than the Dutch had, and turned the toy into science, the Sidereus Nuncius of sixteen ten announcing mountains on the Moon and planets around Jupiter in forty pages that made him the most famous natural philosopher in Europe. The third act is the celebrity and the martyr: the Medici court's philosopher, the defender of Copernicus, the polemicist whose rhetoric won the argument and lost the politics, the trial of sixteen thirty-three and the recantation, the house arrest in which the blind old man organized his life's physics into the Discourses that Newton would inherit. The reader who finishes the volume will hold the whole arc, and will understand why the series gives this one man a shelf of his own: the modern observatory begins here, with a tube of glass and a stubborn mind.

A reading note completes the introduction: the volume's evidence is the man's own paper trail, unusually rich for the century. The letters, the notebooks, the books, and the trial's records survive in the Florentine archives, the manuscripts having been the scholars' four-century quarry, the biography being written from the subject's own hand more fully than any earlier life in this series; the quotations are the man's own words translated plainly, the legends are labeled as legends, and the disputed episodes carry the scholars' disagreements rather than a settled verdict, the volume's honesty being the archive's own, the introduction's last promise being the shelf's standard one kept in the biography's register: the reader finishes the book holding not a monument but a man.

One last introduction card is the series' own family tree, the volume's place on the shelf being the reader's orientation. The Famous Astronomers series holding the lives, this volume's subject being the first of them, the earliest and the largest; the man's having been the founding case, the telescope's inventor-in-use and the method's demonstrator, the later astronomers' being his heirs by the line the chapters trace; and the reader's holding the series' origin story here, the man's life being the trunk from which the later biographies branch, the introduction's map being the shelf's own genealogy, the volume's being the family's eldest, the reader starting the series of lives at its source, the sky's whole modern population of observers descending from one stubborn man with a tube.

Chapter 1: The Discovery of Galileo Galilei

The discovery of Galileo, in the series' sense, is the discovery of what his life discovered, and the first layer is the man's own formation. He was born in Pisa in fifteen sixty-four, the first of six or seven children of Vincenzo Galilei, a lutenist and music theorist of genuine originality, a household in which harmony was debated at the dinner table and authority's arguments were weighed against experiment's evidence; the father taught the son the family trade of doubting the received texts, Vincenzo's own treatises on music having attacked the ancient tuning doctrines by testing them against the sounding string. The son's education followed the pattern of the talented poor: the monastery school at Vallombrosa, the medical degree attempted at Pisa and abandoned for mathematics, the family's finances failing, and the young man's mathematical talent carrying him through, the private tutoring and the small inventions paying the bills until the chairs came.

The first chair, at Pisa, gave him the title and the quarrel. The university's physics was Aristotle's, the doctrine that

heavy bodies fall faster than light ones in proportion to their weight, the doctrine being the textbook of every European faculty; the young mathematician, by the story his own late accounts tell, questioned the doctrine by the leaning tower's demonstrations and by his own reasoning, the anecdote of the tower having its truth in the method rather than the event: the young Galileo had already begun to think of motion as measurable, of falling bodies as a problem for geometry, of the textbook's authority as a hypothesis the world could test. The Pisa years produced the small treatise on motion, the De Motu, unpublished in his lifetime, the young professor already holding the physics he would publish forty years later.

The Padua years, fifteen ninety-two to sixteen ten, are the biography's quiet heart, the eighteen years the man himself called the best of his life. The chair of mathematics at Padua was the Venetian Republic's appointment, the salary better and the atmosphere freer, the university's Protestant-tolerant and papal-skeptical traditions giving the mathematician room the church-run faculties lacked; the private students and the instrument trade supplemented the income, the proportional compass, the military and geometrical instrument he manufactured and taught, being a genuine business with a workshop and a reprintable manual; the household formed around Marina Gamba, the Venetian who bore him three children, the two daughters and the son, the daughters later placed in convents and the son later legitimized, the family's shape being the century's own pragmatic arrangement. The physics of the Padua years is the legend's substance: the falling-body question pursued by inclined-plane experiments, the pendulum's isochronism studied and its law abstracted, the motion's laws being worked out in notebooks and demonstrated in lectures, the whole of the new science of motion being assembled a decade before the telescope gave him a new subject.

The telescope's arrival in sixteen eight or nine is the discovery chapter's hinge. The Dutch spyglass, the invention of spectacle-makers in Holland, arrived in Paris and in Venice as a toy and a military secret, the States-General having considered the device's monopoly value; Galileo heard of it in Padua, never having seen the Dutch instrument, and worked out from the optics alone, by reasoning, what the device must be: a plano-convex objective and a concave eyepiece at the right separation, the tube's length setting the magnification. He built the instrument, improved it beyond the Dutch three-power, and presented an eight-power to the Venetian Senate as a gift and a salary argument, the telescope's military value to the sea republic being the mathematician's leverage; the senate doubled his salary for life, and the Paduan professor had, at forty-five, the security and the instrument that would change the sky.

The transformation from toy to science is the discovery the volume records. In the winter of sixteen nine and ten, Galileo built better tubes, twenty and thirty power, and did what no one else had thought to do: he aimed them above the horizon, at the Moon and the stars rather than at ships and enemies. The Moon showed a rough surface, the terminator's line revealing mountains and craters by their shadows, the arithmetic of the shadows giving the peaks' heights, the Aristotelian doctrine of the perfect celestial spheres dying in a month of evenings; the Milky Way resolved into stars, the nebulae into clusters, the sky's population being suddenly uncountable; and in January sixteen ten the four small stars appeared near Jupiter, moving nightly, the pattern of their motion resolving into orbits around the planet, four moons where the cosmology had allowed none. The Sidereus Nuncius, the Starry Messenger, published in Venice in March sixteen ten, reported the whole harvest in the mathematician's plain style, and the book made its author, in one season, the most famous natural philosopher alive, the discovery of Galileo being complete: the man had become the instrument's master, and the sky had become a place.

The discovery narrative's Roman chapter belongs here as well, the sixteen eleven journey that the fame required. The Sidereus having reached the Jesuits' college, Clavius's astronomers having verified the Jupiter moons with their own tubes, the mathematician traveled to Rome to show the sky to the cardinals, the presentations having been the celebrity's own season, the strangers having looked through the tube and some having seen and some having refused; the Lyncean Academy's prince having made him the first member, the title page's constellation being the honor's own mark; and the reception's mixed verdict being the pattern the whole career would follow, the observations accepted by the working astronomers and resisted by the philosophers whose system they broke, the journey being the fame's first cost and first dividend, the man having left Padua a professor and returned a phenomenon.

The chapter's family register deserves its full page, the household being the man's own foundation. Vincenzo having been the theorist of the lute's strings, the dissonance treatise having argued the ear's authority against the ancient numbers, the father's music having been the son's first lesson in measurable nature; the mother Giulia having been the difficult temperament's other source, the household quarrels being the son's inheritance of tongue; the siblings having been the economist's burden, the younger brothers being the professor's dependents and the

sister's dowry being the Paduan debts' cause, the family being the career's own gravity, the register closing with the observation the biographers repeat: the scientist's formation being the musician's household, the scales and the strings being the childhood's own physics, the man's ear having been the father's gift and the man's tongue having been the family's own weapon.

The chapter's Paduan anatomy deserves one more page, the professor's own working day being the discovery's soil. The lectures having been the Euclid and the Ptolemy the salary required, the sphere's course being the official subject; the private students having been the real teaching, the geometry and the fortification lessons being the foreigners' own curriculum, the military mathematics being the era's applied market; the house having been the workshop's home, the instruments being made in the rooms the family also lived in, the business and the science being one address; and the sixteen-oh-four comets' having been the sky's first public appearance in the career, the nova having been the year's own debate, the man's having published nothing then, the comet's having been the seed the later polemics harvested, the day being the discovery's context, the career's being made of mornings and rent and lens-grinding, the discovery chapter's honest texture.

The chapter closes with the discovery's own economics, the fame's cost and yield being the last ledger. The Sidereus' having sold out its printing in a week, the pamphlet's success having been the publisher's own surprise, the book's having been the career's turning point; the offers having arrived at once, the Pisa chair's having been dangled and the Medici court's having been won, the mathematician's having chosen the prince over the professorate, the choice being the discovery's own gamble; and the salary's having been the instrument's ultimate yield, the tube's having paid better than the sky's truth itself, the chapter's closing observation being the honest economics: the discoveries being free and the discoverer being expensive, the century's having had to learn to fund its geniuses, the man's career being the funding model's first case study.

The chapter adds the confirmation's own record, the later checks the era performed. The English visitors' having carried the tube home, the ambassadors' gifts having been the instrument's own diffusion, the court of James having seen the moons within two years; the French astronomers' having built their own tubes, the Paris observations having confirmed the Medicean stars and extended the record, the moons' tables having been the international science's first shared dataset; and the telescope's having become the standard equipment of the observatories the century then founded, the instrument's own success being the discovery's ratification, the record being the chapter's closing card: the man's claims having been checked by strangers and confirmed by rivals, the confirmation being the discovery's other half, the science's having been public from the cradle, the chapter's last page being the world's own signature under the man's.